

Credit Name: CSE 2140

Assignment Name: Chapter 3 Variables and Constants

Understanding the Problem

How did you approach understanding the challenge: For exercise 6 I knew that I could recycle the code from my digits skillbuilder, and just add the hundreds digit part, and adjust the division. For exercise 5 based off the instructions I knew that I just had to ask for a input and perform basic calculations with the code to generate a answer.

Were there any parts of the problem you found confusing at first? If so, how did you resolve that confusion? When I first added the hundreds place in exercise 6 the division I added didn't work, so I had to use trial and error until I was able to find the correct code to perform the task.

Did you create a plan or break the problem into smaller steps before coding?

Not really, but during my session of coding I gradually broke exercise 5 into different parts, like the quarters, dimes, nickels, and pennies.

How did you decide on the tools, data structures, or algorithms to use?

For exercise 6 I took into account that they could only enter a 3 digit number above 1 so it couldn't physically have a decimal, and when dividing it by ten the program would automatically round so in exercise 6 every variable could be defined as int instead of double or a different tag.

For exercise 5 based on the question using change I initially thought that most of the variables would be defined as double to include decimals/cents. As I gradually started to work out the problem I realized that they asked for the user to input as a whole number in cents for example 98 cent instead of 0.98 dollars. This meant that all of my variables could again be defined as int instead of double.

Did you write the code in small pieces or attempt the entire solution at once?

For the most part I attempted both as a whole project, but as I started to struggle on exercise 5 I broke it up into 4 parts for each coin value.

How did you test your solution along the way to make sure it was working?

For exercise 6 I didn't test along the way but for exercise 5 I tested it after each new coin value.

What part of the problem was the most difficult for you?

Between the 2 exercises the most difficult part was adjusting the equation for the different coin values on exercise 5. I kept trying to just divide the total change by the different amounts, but it realized I had to deduct each previous coin value and then do the next.

How did you handle moments when you felt stuck or unsure of what to do next?

Once I got stuck I went back and reviewed my previous skillbuilders to get inspiration/answers for how to solve my problem.

Was there anything you learned that you think will help you with future challenges?

The main thing that I learned was the significance of the % symbol when dividing out different outcomes from one number. Asking google I learned that it calculates the remainder of a division problem.